

Analytic Sequent Calculus for Coalition Logic with Individual Knowledge

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Motivation

Coalition Logic **CL** (Pauly 2001)

A modal logic designed for reasoning about the ability of groups of agents called **coalitions**.

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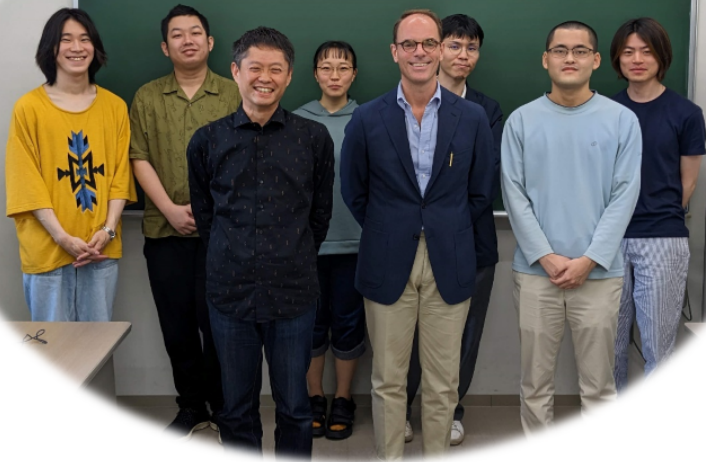
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Coalition Logic with Individual Knowledge **CLK** (Ågotnes and Alechina 2016)

A combination that makes it possible to reason about abilities involving **knowledge** based on **S5**.

*Philosophy (Lecture):
Topics in Modal Logic 2023 (1)/(2)*



Hokkaido Summer Institute 2023

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A combination that makes it possible to reason about abilities involving **knowledge** based on **S5**.

A Gap in Proof-theoretic Studies

There exists sequent calculi for **CL** and **S5**, but **no** sequent calculus for **CLK**.

Outline

- 1 Coalition Logic with Individual Knowledge
- 2 Sequent Calculus and Analytic Cut
- 3 Sequent Calculus $G(\mathbf{CLK})$ for $\mathbf{CLK}$

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“G can make φ come true **no matter what** the other agents do” (Ågotnes and Alechina 2016)

“G is **effective** for φ ” (Pauly 2001)

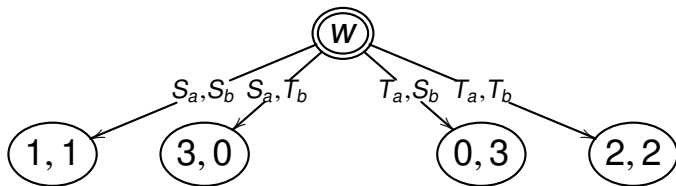
Prisoner's Dilemma

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$a \backslash b$	Silent	Testify
Silent	(1 year; 1 year)	(3 years; 0 year)
Testify	(0 year; 3 years)	(2 years; 2 years)

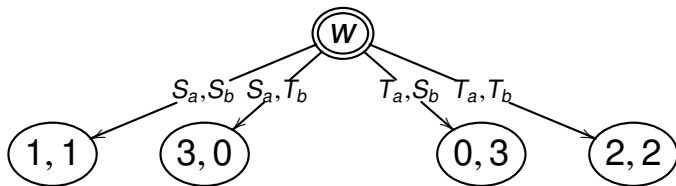
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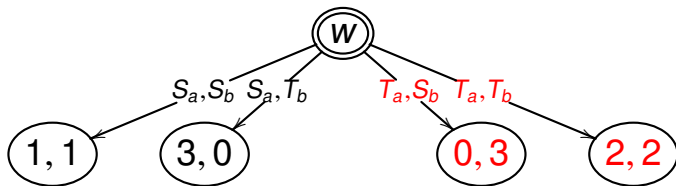


- p : b receives at least one year in prison.

$$M, w \models [\{a\}]p$$

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Adding Individual Knowledge

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$$\Box_a \varphi \rightarrow [\{a\}] \Box_b \varphi$$

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“a can **convey** his knowledge of φ to b .”

Syntax

$$L^- \ni \varphi ::= p \mid \perp \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \rightarrow \psi \mid [G]\varphi$$

$$L \ni \varphi ::= p \mid \perp \mid \neg\varphi \mid \varphi \wedge \psi \mid \varphi \rightarrow \psi \mid [G]\varphi \mid \Box_a\varphi$$

where $G \subseteq \text{Ag}$.

Effectivity Function

$$E : W \rightarrow \mathcal{P}(\mathcal{P}(W))^{\mathcal{P}(\text{Ag})}$$

that maps a tuple $(w, G) \in W \times \mathcal{P}(\text{Ag})$ to a set of sets of states $E(w)(G) \subseteq \mathcal{P}(W)$.

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- $X \in E(w)(G)$ means that coalition G has the **choice** of X , where X is the **set of outcomes**.

Playability (Pauly 2001, 2002)

- **Liveness:** $\emptyset \notin E(w)(G)$
- **Safety:** $W \in E(w)(G)$
- **Ag-maximality:**
 $\overline{X} \notin E(w)(\emptyset) \rightarrow X \in E(w)(\text{Ag})$
- **Outcome Monotonicity:**
 $\forall X \subseteq Y \subseteq W (X \in E(w)(G) \rightarrow Y \in E(w)(G))$
- **Superadditivity:**

$$G \cap H = \emptyset \wedge X \in E(w)(G) \wedge Y \in E(w)(H) \\ \rightarrow X \cap Y \in E(w)(G \cup H)$$

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Frames, Models and Semantics

- An **epistemic coalition frame** F is a tuple $(W, E, (R_a)_{a \in \text{Ag}})$ where (W, E) is a **coalition frame** and R_a is an **equivalence** relation on W for each $a \in \text{Ag}$.

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- An **epistemic coalition model** M is a tuple (F, V) where F is an epistemic coalition frame and $V : \text{Prop} \rightarrow \mathcal{P}(W)$.
- $\mathbf{F}_{\text{CLK}}$: the class of all epistemic coalition frames.

Semantics

Let $M = (W, E, V)$ be a coalition model.

$M, w \models \Box_a \varphi$ iff for all $v \in W$,
 $wR_a v$ implies $M, v \models \varphi$,

$M, w \models [G]\varphi$ iff $\llbracket \varphi \rrbracket_M \in E(w)(G)$,

where $\llbracket \varphi \rrbracket_M = \{ w \in W \mid M, w \models \varphi \}$.

Axiomatization H(**CLK**) for **CLK**

(Taut)	all instances of propositional tautologies
(G $\perp$)	$\neg[G]\perp$
(G $\top$)	$[G]\top$
(Agmax)	$\neg[\emptyset]\neg\varphi \leftrightarrow [Ag]\varphi$
(G $\wedge$)	$[G](\varphi \wedge \psi) \rightarrow [G]\varphi$
(Superadd)	$[G]\varphi \wedge [H]\psi \rightarrow [G \cup H](\varphi \wedge \psi)$, where $G \cap H = \emptyset$
(MP)	From $\varphi \rightarrow \psi$ and φ infer ψ
(RG)	From $\varphi \leftrightarrow \psi$ infer $[G]\varphi \leftrightarrow [G]\psi$
(K)	$\Box_a(\varphi \rightarrow \psi) \rightarrow (\Box_a\varphi \rightarrow \Box_a\psi)$
(T)	$\Box_a\varphi \rightarrow \varphi$
(5)	$\neg\Box_a\varphi \rightarrow \Box_a\neg\Box_a\varphi$
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Axiomatization $H(\mathbf{CLK})$ for $\mathbf{CLK}$

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Fact (Ågotnes and Alechina 2016)

$H(\mathbf{CLK})$ is sound and complete for $\mathbb{F}_{\mathbf{CLK}}$.

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Gerhard Gentzen

Untersuchungen über das logische Schließen*). I.

Von

Gerhard Gentzen in Göttingen.

Übersicht.

Die folgenden Untersuchungen beziehen sich auf den Bereich der Prädikatenlogik [bei H.-A.¹⁾ „engerer Funktionenkalkül“ genannt]. Diese umfaßt solche Schlüsse, die in allen Teilen der Mathematik immerzu gebraucht werden. Was noch zu ihnen hinzukommt, sind Axiome und Schlußweisen, die man den einzelnen Zweigen der Mathematik selbst zurechnen kann, z. B.

G. Gentzen (1934). "Untersuchungen über das logische Schließen. I".
Mathematische Zeitschrift. 39 (2): 176–210.

Sequent Calculi for Modal Logics

$$\varphi_1, \dots, \varphi_m \Rightarrow \psi_1, \dots, \psi_n$$

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$$\Gamma \Rightarrow \Delta$$

$$\bigwedge \Gamma \text{ implies } \bigvee \Delta$$

Sequent Calculi for Modal Logics

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Sequent Calculus for **CL**

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- Sequent calculus $G(\mathbf{CL})$ for **CL** by Schröder and Pattinson (2008):

$$\frac{\varphi_1, \dots, \varphi_m \Rightarrow \Psi, \gamma}{[G_1]\varphi_1, \dots, [G_m]\varphi_m \Rightarrow [Ag]\Psi, [H]\gamma} ([\cdot])^\dagger$$

$$\frac{\varphi_1, \dots, \varphi_m \Rightarrow}{[G_1]\varphi_1, \dots, [G_m]\varphi_m \Rightarrow} ([\cdot]_D)^\dagger.$$

$$\dagger : G_i \cap G_j = \emptyset \ (i \neq j) \text{ and } G_i \subseteq H.$$

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Fact (Schröder and Pattinson 2008)

The (*Cut*) rule is **admissible** in $G(\mathbf{CL})$.

What's New on This Paper?

Main Result of This Talk

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- 1 $G(\mathbf{CLK})$: **not** cut-free, but enjoys the subformula property (i.e., analytic) as in (Takano 1992).

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Main Result of This Talk

- 1 $G(\mathbf{CLK})$: **not** cut-free, but enjoys the subformula property (i.e., analytic) as in (Takano 1992).
- 2 Corollary: Craig Interpolation Property of $\mathbf{CLK}$
If $G(\mathbf{CLK}) \vdash \varphi \rightarrow \psi$, then there exists a formula χ satisfying the following conditions:
 - a $G(\mathbf{CLK}) \vdash \varphi \rightarrow \chi$ and $G(\mathbf{CLK}) \vdash \chi \rightarrow \psi$;
 - b $\text{Prop}(\chi) \subseteq \text{Prop}(\varphi) \cap \text{Prop}(\psi)$.

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Sequent Calculus G(**CLK**)

- Rules from G(**S5**):

$$\frac{\Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi}{\Box_a \Gamma \Rightarrow \Box_a \Delta, \Box_a \varphi} (\Box \mathbf{S5}) \quad \frac{\varphi, \Gamma \Rightarrow \Delta}{\Box_a \varphi, \Gamma \Rightarrow \Delta} (\Box \Rightarrow).$$

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- Rules from G(**CL**):

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Proof Example

- **Superadditivity:**

$$\begin{aligned} G \cap H = \emptyset \wedge X \in E(w)(G) \wedge Y \in E(w)(H) \\ \rightarrow X \cap Y \in E(w)(G \cup H) \end{aligned}$$

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- (Superadd):

$$[G]\varphi \wedge [H]\psi \rightarrow [G \cup H](\varphi \wedge \psi),$$

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$$\frac{\frac{\varphi \Rightarrow \varphi}{\varphi, \psi \Rightarrow \varphi} (w \Rightarrow) \quad \frac{\psi \Rightarrow \psi}{\varphi, \psi \Rightarrow \psi} (w \Rightarrow)}{\varphi, \psi \Rightarrow \varphi \wedge \psi} (\Rightarrow \wedge)$$

$$\frac{\varphi, \psi \Rightarrow \varphi \wedge \psi}{[G]\varphi, [H]\psi \Rightarrow [G \cup H](\varphi \wedge \psi)} ([\cdot])$$

$$\vdots$$

Equipollence

Proposition

- 1 If $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{CLK})$ then $\bigwedge \Gamma \rightarrow \bigvee \Delta$ is a theorem of $H(\mathbf{CLK})$.
- 2 If φ is a theorem of $H(\mathbf{CLK})$, then $\Rightarrow \varphi$ is derivable in $G(\mathbf{CLK})$.

Sequent Calculus $G(\mathbf{CLK})$

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The (*Cut*) rule is **admissible** in $G(\mathbf{CL})$.

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The (*Cut*) rule is **admissible** in G(**CL**).

However...

$$\frac{\frac{\frac{\Box_a \neg p \Rightarrow \Box_a \neg p}{\Rightarrow \neg \Box_a \neg p, \Box_a \neg p} (\Rightarrow \neg) \quad \frac{\frac{p \Rightarrow p}{\neg p, p \Rightarrow} (\neg \Rightarrow) \quad \frac{\Box_a \neg p, p \Rightarrow}{(K \Rightarrow)} \quad \frac{\Box_a \neg p, p \Rightarrow}{(Cut) \Box_a \neg p.}}{\Rightarrow \Box_a \neg \Box_a \neg p, \Box_a \neg p} (\Rightarrow \Box_{S5})}{p \Rightarrow \Box_a \neg \Box_a \neg p}$$

Sequent Calculus G(**CLK**)

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Fact (Ohnishi and Matsumoto 1959)

(B) is not derivable in G(**S5**) w/o (*Cut*).

Sequent Calculus G(**CLK**)

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(B) is not derivable in G(**CLK**) w/o (*Cut*).

The (*Cut*) rule is **not** admissible in G(**CLK**)!

Sequent Calculus G(**CLK**)

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Subformula Property of G(**S5**) (Takano 1992)

If $\Gamma \Rightarrow \Delta$ is derivable in G(**S5**) then it is derivable in G(**S5**) with only analytic applications of (*Cut*).

Analytic Cut in G(**CLK**)

- A cut

$$\frac{\Gamma \Rightarrow \Delta, \varphi \quad \varphi, \Pi \Rightarrow \Sigma}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut)_\varphi$$

is **analytic** if $\varphi \in \text{Sub}(\Gamma, \Pi, \Delta, \Sigma)$.

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- Define **G^a(CLK)** as the same calculus as G(**CLK**) except that (**Cut**) is always analytic.

Analytic Cut Property of $G(\mathbf{CLK})$

- If a sequent is derivable in $G(\mathbf{CLK})$,
then it is derivable in $G^a(\mathbf{CLK})$.
- If $\Gamma \Rightarrow \Delta$ is derivable in $G(\mathbf{CLK})$, then it is derivable in $G(\mathbf{CLK})$ by a derivation which consists of $\text{Sub}(\Gamma, \Delta)$ alone.

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(Cor.) Consistency of $G(\mathbf{CLK})$

The sequent $\Rightarrow$ is **not** derivable in $G(\mathbf{CLK})$.

Non-analytic Cut Elimination of $G(\mathbf{CLK})$

- Goal: if a sequent is derivable in $G(\mathbf{CLK})$ then it is derivable in $G^a(\mathbf{CLK})$.

Non-analytic Cut Elimination of $G(\mathbf{CLK})$

- Goal: if a sequent is derivable in $G(\mathbf{CLK})$ then it is derivable in $G^a(\mathbf{CLK})$.
- We use the following lemma which is modified from Takano (1992)'s argument

Key Lemma (proof is very involved)

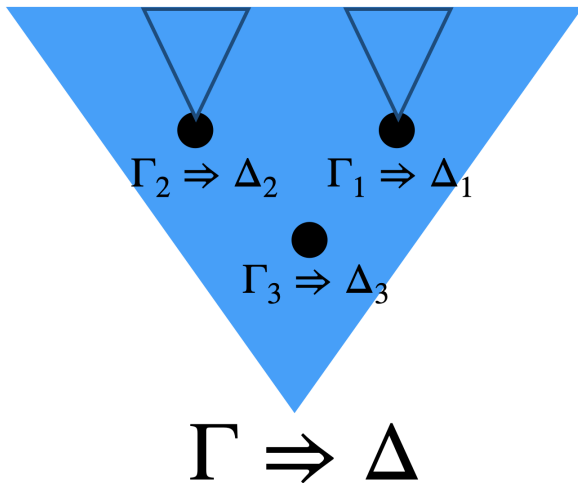
If

$$G^a(\mathbf{CLK}) \vdash \Box_a \Gamma \Rightarrow \Box_a \Delta, (\Box_a \varphi)^m$$

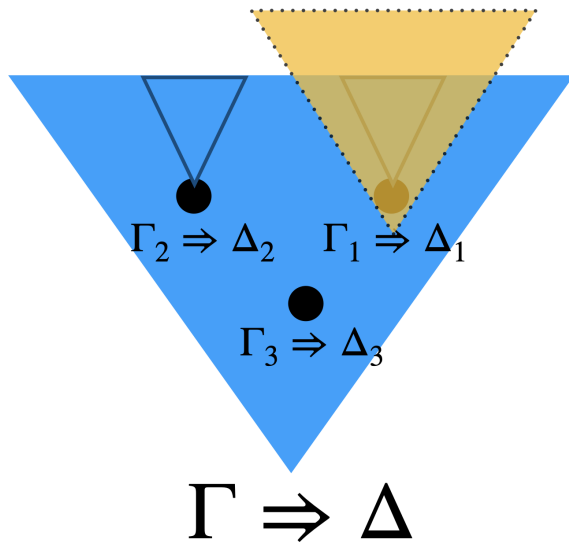
with an application of $(\Box_{S5})$ for its lowest inference, then

$$G^a(\mathbf{CLK}) \vdash \Box_a \Gamma \Rightarrow \Box_a \Delta, \varphi.$$

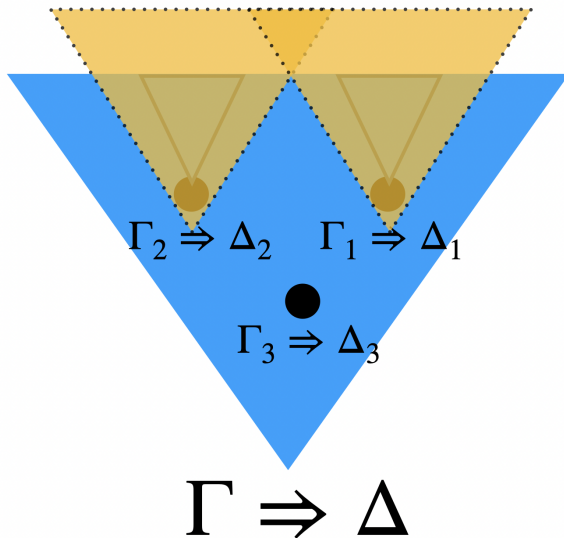
How to Eliminate Non-analytic Cut



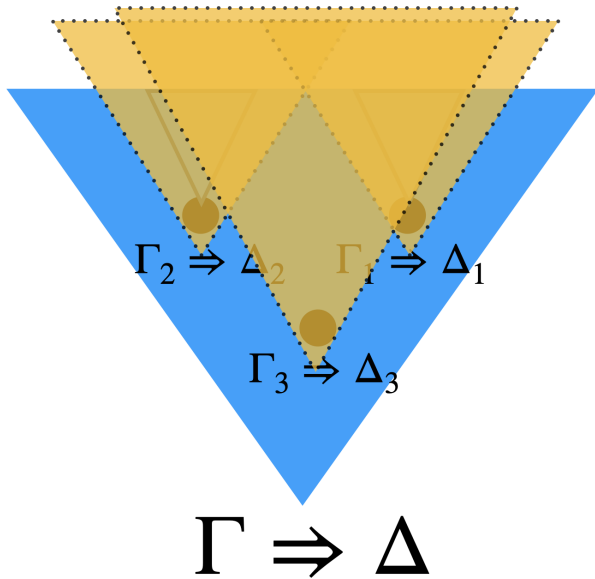
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How to Eliminate Non-analytic Cut

So, it suffices to eliminate the **final** applications of (*Cut*):

$$\mathcal{D} \equiv \frac{\begin{array}{c} \vdots \mathcal{L} \\ \Gamma \Rightarrow \Delta, \varphi \end{array} \quad \begin{array}{c} \vdots \mathcal{R} \\ \varphi, \Pi \Rightarrow \Sigma \end{array}}{\Gamma, \Pi \Rightarrow \Delta, \Sigma} (Cut).$$

Craig Interpolation Property of **CL**

CL (Schröder and Pattinson 2008)

Suppose $G(\mathbf{CL}) \vdash \Rightarrow \varphi \rightarrow \psi$. Then there exists a formula χ such that

- 1 $G(\mathbf{CL}) \vdash \Rightarrow \varphi \rightarrow \chi$ and $G(\mathbf{CL}) \vdash \Rightarrow \chi \rightarrow \psi$;
- 2 $\text{Prop}(\chi) \subseteq \text{Prop}(\varphi) \cap \text{Prop}(\psi)$,

where $\text{Prop}(\varphi)$ denotes the set of all propositional variables appearing in φ .

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Proof Method (Maehara 1961)

By induction on **cut-free** proofs in $G(\mathbf{CL})$.

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- 1 $G(\mathbf{CLK}) \vdash \Rightarrow \varphi \rightarrow \chi$ and $G(\mathbf{CLK}) \vdash \Rightarrow \chi \rightarrow \psi$;
- 2 $\text{Prop}(\chi) \subseteq \text{Prop}(\varphi) \cap \text{Prop}(\psi)$,

where $\text{Prop}(\varphi)$ denotes the set of all propositional variables appearing in φ .

Maehara's Method for a-Cut (Ono 1998)

By induction on proofs with **only** analytic applications of (*Cut*), in which the case for analytic (*Cut*) is discussed in detail .

Conclusion

Main Result of This Talk

➊ $G(\mathbf{CLK})$: analytic.

Conclusion

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- 1 $G(\mathbf{CLK})$: analytic.
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- 1 $G(\mathbf{CLK})$: analytic.
- 2 (Corollary) CIP for $G(\mathbf{CLK})$.
- 3 (Corollary) $G(\mathbf{CLK})$ is decidable.

Conclusion

Further Directions

- Counterexample for the condition of agents of CIP:

$$[\{a\}]p, [\{b\}]\neg p \Rightarrow$$

- Sequent calculus $G(\mathbf{CLK}^+)$ for $\mathbf{CLK}^+$ with the following axiom:

$$[\{a\}]\varphi \rightarrow \Box_a[\{a\}]\varphi.$$

- Sequent calculi for $\mathbf{CL}$ with distributed knowledge $\mathbf{CLD}$ and $\mathbf{CLD}^+$.